

IUI Evaluation Report

Duolingo — Web Onboarding, Placement Test & Lesson Flow

Domain	Education / EdTech
Audience	Adult self-directed language learners using Duolingo's free web product
Persona	Aisha, 29, marketing coordinator learning French casually for an upcoming trip — motivated by streaks and quick wins, easily discouraged if the app doesn't already know what she's capable of
Input	16 screenshots — language picker, placement test, lesson flow, dashboard
Evaluated by	IUI Evaluator v1.1 — literature-grounded rubric
Date	14 August 2026

Score

19/35 (54%)

Band

Developing IUI

IUI Maturity Band

Band	Raw	%
Pre-intelligent	0-10	0-28%
Emerging IUI	11-18	31-51%
Developing IUI	19-25	54-71%
Mature IUI	26-31	74-89%
Exemplary IUI	32-35	91-100%

Duolingo sits right at the entry threshold of Developing IUI: real adaptive and context-aware mechanisms are present and functioning, but the system explains almost none of its reasoning to the user, which caps it well below Mature.

Dimension Scores

#	Dimension	Score	Key Evidence
1	Adaptivity & Personalisation	3/5	Placement test escalates difficulty (confirmed EN→FR direction reversal forcing recall); self-report question's effect on placement is unconfirmed
2	User Modelling & Representation	3/5	Explicit state tracking with visible effect — score meter (10/11) determines starting unit — but the model isn't inspectable
3	Context-Awareness	3/5	Temporal (day-of-week streak calendar) plus workflow state (section/unit progress, daily quests)
4	Transparency & Explainability	1/5	"Let's make this a bit harder..." is the only rationale ever given for an adaptive change

5	User Control & Controllability	3/5	User can opt out of adaptive placement entirely; “Too Easy / Too Difficult” flags feed back into the system
6	Naturalness & Interaction Quality	3/5	Multimodal: speech recognition, audio playback, matching, word-bank construction
7	Proactivity & Task Support	3/5	Daily quests, streak-loss warnings, guidebook prompts — mostly retention-driven
Total		19/35 (54%)	

What This Interface Does Well

The placement flow is genuinely adaptive rather than cosmetic. The “Find my level” path runs a real diagnostic — a score meter that fills as questions are answered, escalating difficulty mid-test (confirmed: the direction of translation flips from French→English to English→French, forcing active vocabulary recall instead of passive recognition), and lands the user in a specific section and unit rather than a generic starting point. That’s a legitimate user model with a visible downstream effect, which is more than most “personalised onboarding” flows manage.

The interface also respects user agency at the point of adaptation: “Start from scratch” sits as an equally-weighted alternative to “Find my level,” so a user who distrusts or dislikes the adaptive placement can opt straight out of it. Inside lessons, the “Too Easy / Too Difficult / Report” row on every completed exercise gives Duolingo a continuous, low-friction channel for users to contest the system’s difficulty calibration — a controllability mechanism that’s easy to miss but does real work.

Context-awareness is handled with more nuance than is typical for a habit-loop app. The streak calendar is temporal (day-of-week, “starts tomorrow” framing) rather than a bare counter, and the Daily Quests panel ties session-level progress (“Earn 10 XP — 10/10”) to a persistent task state visible on the dashboard. Combined with the speaking exercise (microphone input) and audio playback throughout, the interaction model is meaningfully multimodal for a language-learning context.

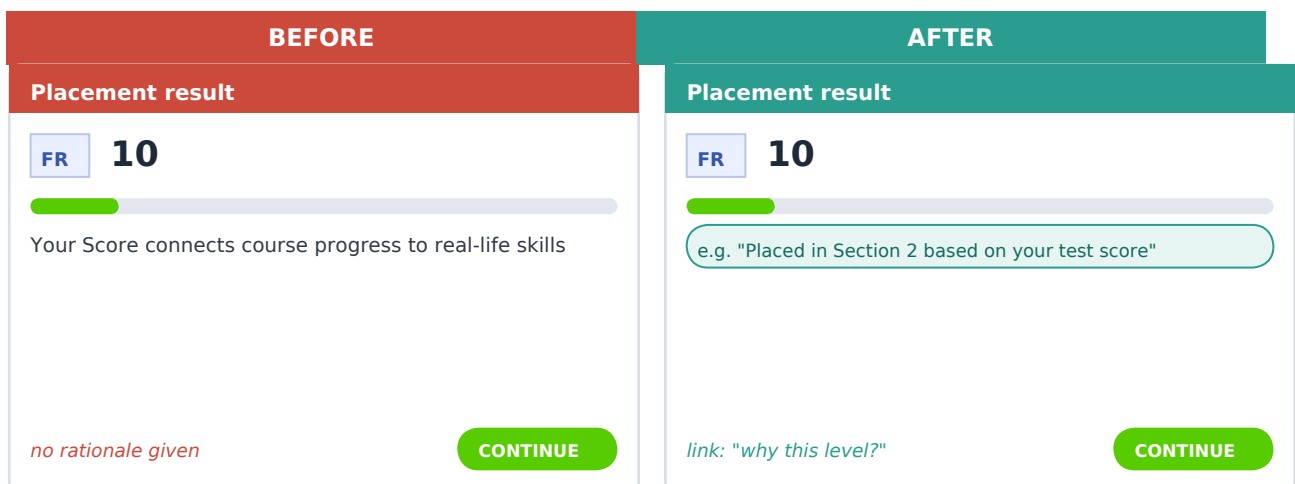
Priority Improvements (Top 3)

Improvement 1: Explain what the placement test actually measured

Dimension: Transparency & Explainability

Why it matters for Aisha: She answers “How much French do you know?” and then sits through an adaptive test, but the interface never tells her which one determined her placement, or why Section 2 specifically was chosen. If she’d expected to land higher, she has no way to understand — let alone contest — the decision.

Suggested approach: After the placement result screen, add a one-line rationale — e.g. “Placed in Section 2 based on your test score” — and if the self-report answer is used at all, say so explicitly. IUI literature basis: Höök (1999) and Mezhoudi et al. (2015) identify transparency as a paramount usability criterion for adaptive systems — unexplained adaptation is experienced as arbitrary rather than intelligent.



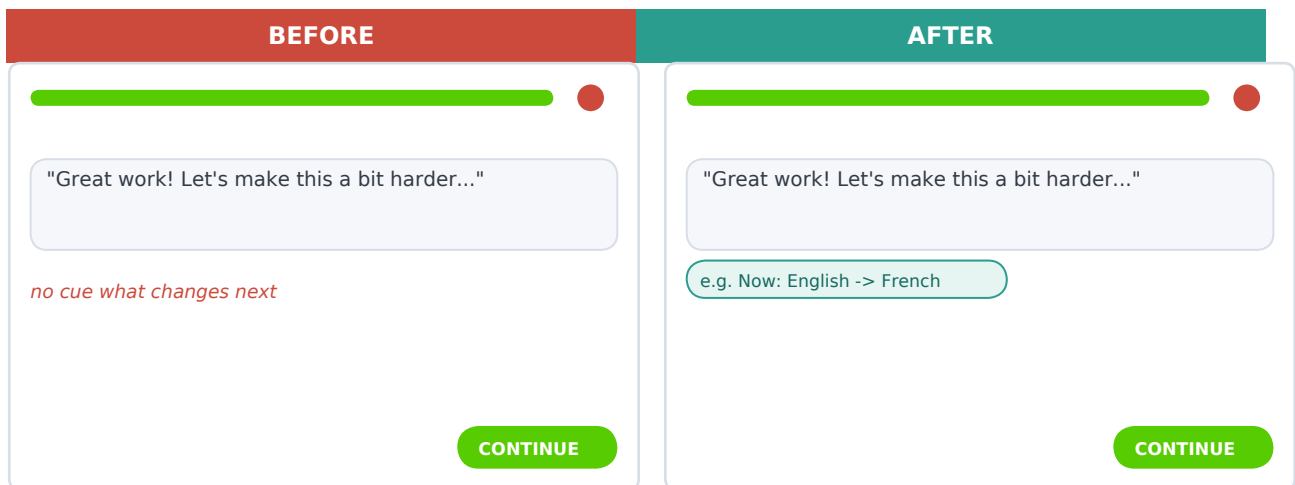
Improvement 2: Surface the difficulty escalation instead of leaving it implicit

Dimension: Transparency & Explainability / Adaptivity

Why it matters for Aisha: The test does escalate meaningfully (confirmed: it shifts from recognition to recall), but “Let’s make this a bit harder...” gives no indication of how. A learner who suddenly struggles can’t tell whether it’s her recall or the format that changed.

Suggested approach: Pair the escalation message with a concrete, honest cue — e.g. a small “Now translating into French” tag on the question header — so the format shift is visible, not just implied.

IUI literature basis: Findlater & Gajos (2009) note that adaptive mechanisms can improve one dimension (test accuracy) while raising cognitive load along another (user confusion) if the adaptation isn’t legible in the moment.



Improvement 3: Give the self-reported proficiency question a visible purpose

Dimension: User Modelling & Representation

Why it matters for Aisha: Asking “How much French do you know?” implies the system is building a model of her, but nothing afterward shows that signal being used — the adaptive test appears to run identically regardless. Unused prompts erode trust in the app’s model even when the rest of it works.

Suggested approach: Either visibly blend the self-report into the diagnostic (e.g. skip fewer adaptive questions for someone who selects “I can have basic conversations”) or, if it’s a segmentation input only, don’t present it inside a flow that otherwise reads as diagnostic.

IUI literature basis: Brdník, Heričko & Šumák (2022) identify representation — the system’s model of the user being both accurate and legible — as one of the three core IUI characteristics.

BEFORE	AFTER
How much French do you know? I'm new to French I know some common words I can have basic conversations <i>answer's effect: unknown</i>	How much French do you know? I'm new to French I know some common words I can have basic conversations e.g. 'This shortens your placement test'

Domain-Specific Commentary

For EdTech, the rubric weights User Modelling, Adaptivity, and Proactivity most heavily, since the core value proposition is scaffolding a learner's actual skill level rather than a fixed curriculum. Duolingo earns its middle-of-the-pack score here: the adaptive placement mechanism (Adaptivity: 3, User Modelling: 3) is real and consequential, but Proactivity (3) leans heavily toward retention nudges — streak-loss warnings, quest bars, leaderboard-unlock prompts — rather than learning-directed proactivity (e.g. “you’re consistently missing verb conjugations, here’s a focused drill”). For a learner like Aisha whose motivation is streak-driven, that’s not necessarily a design flaw, but it does mean the intelligence on display is more habit-engine than tutor.

Overall Verdict

Duolingo qualifies as a Developing IUI: it has a functioning adaptive core — a real diagnostic that changes question format based on measured performance, a user model with a visible downstream effect on placement, and multiple points where the user can steer or opt out of that adaptation — but it pairs that core with almost no transparency, leaving the user to infer or simply not know why the system did what it did. The gap between Developing and Mature here isn’t more intelligence; it’s explaining the intelligence that already exists.